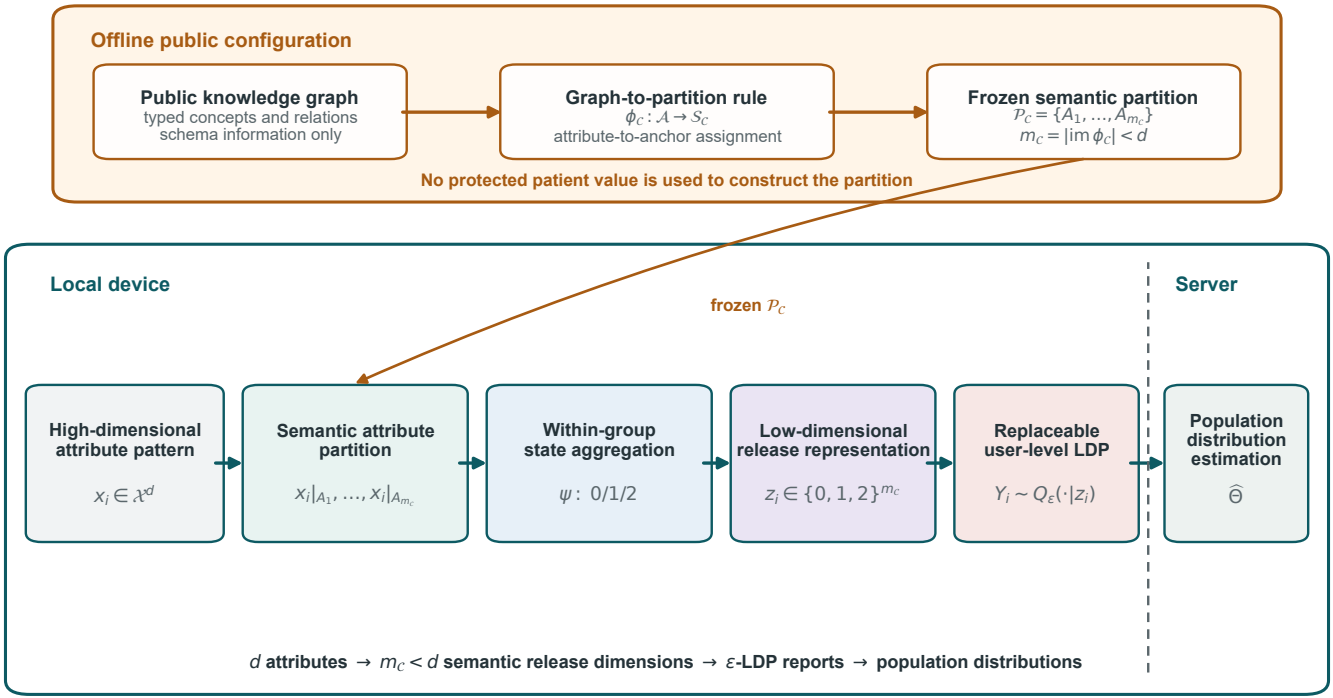

Offline public configuration
Public knowledge graph
typed concepts and relations
schema information only
Graph-to-partition rule
$\phi_C : \mathcal{A} \to S_C$
attribute-to-anchor assignment
Frozen semantic partition
$\mathcal{P}_C = \{A_1, \ldots, A_{m_C}\}$
$m_C = |\mathrm{im}\,\phi_C| < d$
No protected patient value is used to construct the partition
Local device
frozen $\mathcal{P}_C$
Server
High-dimensional attribute pattern
$x_i \in \mathcal{X}^d$
Semantic attribute partition
$x_i|_{A_1}, \ldots, x_i|_{A_{m_C}}$
Within-group state aggregation
ψ: 0/1/2
Low-dimensional release representation
$z_i \in \{0, 1, 2\}^{m_C}$
Replaceable user-level LDP
$Y_i \sim Q_\varepsilon(\cdot|z_i)$
Population distribution estimation
$\widehat{\Theta}$
d attributes → $m_C < d$ semantic release dimensions → ε-LDP reports → population distributions